Project 3
Final Presentation

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BIOS 6624

Research Questions

- Primary Analysis - What is the 10-year survival probability for various risk profiles at ages 40, 50 and 60?

- Secondary Analysis - Are there significant changes in systolic blood pressure and diabetes across the three time periods that would suggest time-varying covariates in future analyses?

Data Overview

- Removed 32 individuals due to reported a stroke prior to the study period
- Right-censored anyone that did not have a stroke or death within the 10-year window

Characteristic	Overall N = 4,402[1]	Males N = 1,930[1]	Females N = 2,472[1]
Stroke within 10 Years	111 (2.5%)	49 (2.5%)	62 (2.5%)
Diabetes			
No	4,283 (97%)	1,871 (97%)	2,412 (98%)
Yes	119 (2.7%)	59 (3.1%)	60 (2.4%)
Systolic Blood Pressure (mmHg)	133 (22)	132 (19)	134 (24)
Age (years)	50 (9)	50 (9)	50 (9)
Coronary Heart Disease			
Free of Disease	4,215 (96%)	1,810 (94%)	2,405 (97%)
Prevalent Disease	187 (4.2%)	120 (6.2%)	67 (2.7%)
Taking Blood Pressure Medication			
Not Currently Taken	4,206 (97%)	1,868 (98%)	2,338 (96%)
Currently Taken	136 (3.1%)	40 (2.1%)	96 (3.9%)
Missing	60	22	38
Smoking Status			
Not Current Smoker	2,233 (51%)	762 (39%)	1,471 (60%)
Current Smoker	2,169 (49%)	1,168 (61%)	1,001 (40%)
Total Cholesterol (mg/dL)	237 (45)	234 (42)	240 (46)
Missing	52	7	45
BMI (kg/m^2)	25.8 (4.1)	26.2 (3.4)	25.6 (4.5)
Missing	17	4	13

[1] n (%); Mean (SD)

Analysis

Primary Analysis - Variable Selection & Cox Proportional Hazard Models

- Backward Selection using AIC to select variables
- Fitted separate Cox PH Models for males and females
  - Required Predictors: Systolic Blood Pressure, Diabetes, Age
  - Potential Predictors: Prevalent Coronary Heart Disease, Blood Pressure Medication, Total Cholesterol, Current Smoking Status, BMI

Secondary Analysis - Descriptive Examination

- Assessed average systolic blood pressure and proportion of diabetes across time periods
- Calculated Proportion of participants with clinically meaningful changes across the time periods

Results - Primary Analysis

	Female			Male		
Predictor	**HR**	**95% CI**	**p-value**	**HR**	**95% CI**	**p-value**
Age	1.068	(1.03, 1.109)	<0.001	1.070	(1.031, 1.11)	<0.001
Systolic Blood Pressure	1.029	(1.021, 1.037)	<0.001	1.032	(1.021, 1.043)	<0.001
Diabetes	1.597	(0.576, 4.429)	0.368	4.287	(1.862, 9.873)	<0.001
Current Smoker	1.315	(0.765, 2.259)	0.322	2.013	(1.084, 3.738)	0.027

Age	Healthy	High BP	Diabetes	High BP + Diabetes	Smoker
Female					
40	0.997	0.986	0.995	0.978	0.995
50	0.993	0.973	0.989	0.957	0.991
60	0.987	0.949	0.980	0.919	0.983
Male					
40	0.997	0.987	0.988	0.944	0.995
50	0.995	0.974	0.977	0.894	0.989
60	0.990	0.950	0.956	0.803	0.979

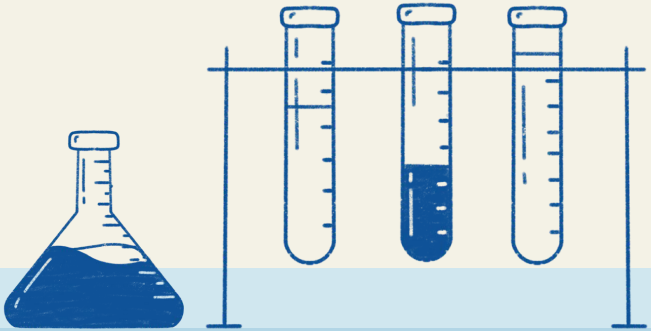
Results - Secondary Analysis

	Period 1	Period 2	Period 3
Female			
SYSBP	131.7 (19.4)	135.5 (19.9)	139.3 (21.2)
Diabetes	3%	4.4%	8.7%
Male			
SYSBP	133.8 (24.5)	138.1 (24.3)	140.9 (24.1)
Diabetes	2.5%	3.6%	7.1%

Percentage of Participants that experienced a clinically meaningful change across the time periods:

- ≥10 mmHg - 69.5%
- Any change - 4.9%

Discussion

- Age and systolic blood pressure were significant predictors of stroke in both sexes. Notably, a substantially stronger effect of diabetes in males (HR = 4.287) compared to females (HR = 1.597),
- The risk profile analysis highlighted that the combined presence of hypertension and diabetes produced the greatest reductions in stroke-free survival, particularly among older males.
- Due to the high number of participants who saw changes in systolic blood pressure greater than 10 mmHg, time-varying covariates are suggested for consideration for future analyses.
- Potential Limitations:
  - All predictors were measured at baseline only, which as demonstrated by the secondary analysis, may not adequately capture dynamic changes over follow-up.
  - Interaction terms between predictors were not considered
  - The Framingham cohort's limited demographic diversity may restrict generalizability.

Questions?